Robed Beauvile-Owen, Ph.D.

Data Scientist | Measurement, Experimentation & Predictive Modeling

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EXPERIENCE

Founder & Data Scientist, *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy that measures demand and opportunity across counties to support site selection, with feasibility screenings that steered clients away from low-potential sites.
- Translated owners' profit goals into required customer-volume targets with a revenue driver tree (traffic × capture rate × average ticket), and quantified each site's downside risk with Monte Carlo simulation (probability of loss, VaR, CVaR).
- Estimated that the average new location adds about $533K in market revenue a year, using Double ML (econml, XGBoost) validated with difference-in-differences and a synthetic-data recovery test.
- Designed a site-scoring model that ranks locations on demographic, traffic, competitive-density, and consumer-behavior features, fed by reusable Python ETL pipelines (250+ unit tests) and reported with explicit assumptions and uncertainty ranges.

Senior Data Scientist, *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Built an in-house experimentation platform that read out creative A/B, champion/challenger, and mail-holdout tests to guide campaign budgets, automating power analysis and significance tests, with CUPED narrowing confidence intervals (20% lower variance).
- Built automated reporting tools for monthly and weekly reporting, metric monitoring, and Power BI dashboards used by data science and strategy.
- Standardized core KPIs (gross lead rate, enrollment rate, cost per member), fixing conflicting lead and attribution rules with data engineering.
- Improved campaign targeting with segmentation models and uplift models trained on mail-holdout tests, using millions of HIPAA-protected (PII/PHI) Medicare/Medicaid-eligible records, adopted in 90% of campaigns.
- Re-architected the production churn prediction pipeline in Python, SQL, and Spark on Databricks, cutting inference runtime 48% and memory 23%.

Founding Machine Learning Engineer, *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- Built predictive pricing models that improved shipping estimates, cutting median error vs. carrier cost from $13 to $4 on standard orders while maintaining contribution margin.
- Deployed a regression pricing API in C# on Azure (10,000+ monthly transactions), trained on SQL tables that split bulk orders into item-level costs.
- Tuned Elasticsearch relevance (custom analyzers, field boosting) to cut search dissatisfaction 12% and raise click-through rate, validated with NDCG@10 and online A/B tests.

Data Scientist, *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Built an XGBoost lead-scoring model on millions of healthcare consumer records with hundreds of external attributes, raising offline AUC-ROC, explaining drivers with SHAP, and confirming a 9% relative GLR lift over the prior model in a randomized champion/challenger mail test.
- Owned the monthly model monitoring report (drift, accuracy, forecast vs. actual) for up to 5 production models, read by the managing partner.
- Doubled the precision of high-opportunity zone selection (15% to 30%, measured against past successful campaigns) with GeoPandas spatial analysis.

TECHNICAL PROJECTS

Medicare Claims & Fraud Detection github.com/beauvilerobed/.../projects/medicare-claims-intelligence

- Built a dbt warehouse with 26 data tests over 790K synthetic CMS Medicare claims and a fraud (FWA) detector checked against planted billing schemes.
- Built an explainable XGBoost model (SHAP) that finds 46% more next-year high-cost members than ranking by past cost.

Measuring What Works: Economics in Python github.com/beauvilerobed/.../projects/causal-economics-python

- Measured which groups benefit most from a randomized cash transfer experiment (Indonesia's PKH) by estimating heterogeneous treatment effects with GATES and causal forests.
- Measured the gender wage gap net of worker and job characteristics in 2018 CPS data, using Double ML to separate the gap from confounders.

CaLaun - Univariate Calculus Lab (Math Tutor & AI Assistant) github.com/calaun-learning/calaun

- Built a symbolic math engine (SymPy) that shows deterministic step-by-step solutions for derivatives, integrals, and limits.
- Added a self-hosted Llama 3.2 assistant (Ollama) that reads the current solution and explains each step.

EDUCATION

Ph.D. in Mathematics, *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics, *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Measurement: KPI Frameworks, Driver Trees, Leading Indicators, Forecasting, Monte Carlo Simulation (VaR, CVaR), Model & Drift Monitoring

Experimentation & Causal Inference: A/B Testing (Creative, Champion/Challenger, Holdout), Power Analysis, CUPED, Uplift Modeling, Double ML, DiD

Predictive Modeling: Regression, Classification, Segmentation, XGBoost, Survival Analysis, SHAP, LLMs

Tools: Python (pandas, scikit-learn, statsmodels, econml, GeoPandas), SQL (Snowflake, SQL Server), Spark, Databricks, dbt, C#, Azure, Power BI, Tableau